

I. Sampling Distributions

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i = 1, \dots, n$$

Under classical SLR assumptions:

$$y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{SSX}\right)\right)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{SSX}\right)$$

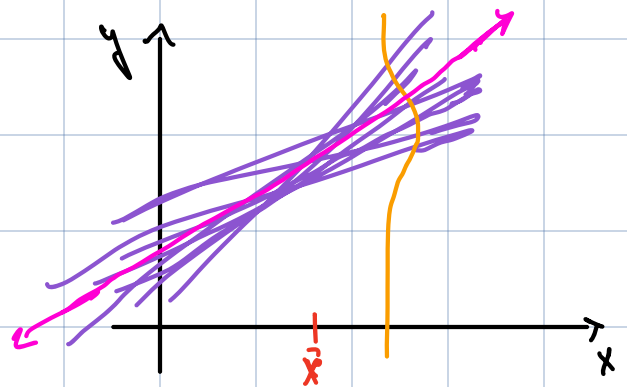
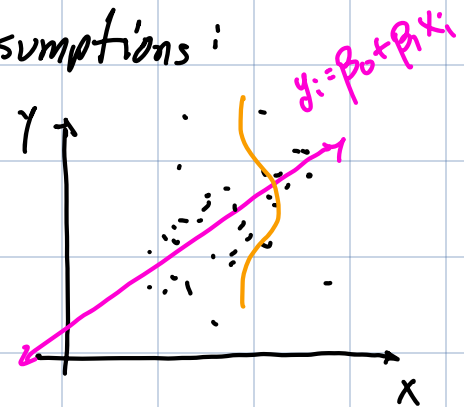
$$\text{where } SSX = \sum_i (x_i - \bar{x})^2$$

Fitted values $\hat{y}_i$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

What is the dist'n of $\hat{y}_i$?

$$\hat{y}_i \sim N(E(\hat{y}_i), \text{Var}(\hat{y}_i))$$



$$E(\hat{y}_i) = E(\hat{\beta}_0 + \hat{\beta}_1 x_i) = E(\hat{\beta}_0) + E(\hat{\beta}_1) x_i \\ = \beta_0 + \beta_1 x_i$$

$$\text{Var}(\hat{y}_i) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

Recall $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$
 But, we don't need this

$$= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i)$$

$$= \text{Var}\left(\frac{1}{n} \sum_j y_j + \hat{\beta}_1 (x_i - \bar{x})\right)$$

$$= \text{Var}\left(\frac{1}{n} \sum_j y_j + (x_i - \bar{x}) \sum_j k_j y_j\right)$$

Recall

$$\hat{\beta}_1 = \sum_j k_j y_j$$

$$= \text{Var}\left(\sum_j \underbrace{\left(\frac{1}{n} + (x_i - \bar{x}) k_j\right)}_{\tilde{k}_j} y_j\right)$$

$$= \text{Var}\left(\sum_j \tilde{k}_j y_j\right)$$

$$= \sum_j \text{Var}(\tilde{k}_j y_j)$$

$$= \sum_j \tilde{K}_j^2 \text{Var}(y_{ij})$$

$$= \sigma^2 \sum_j \tilde{K}_j^2$$

⋮

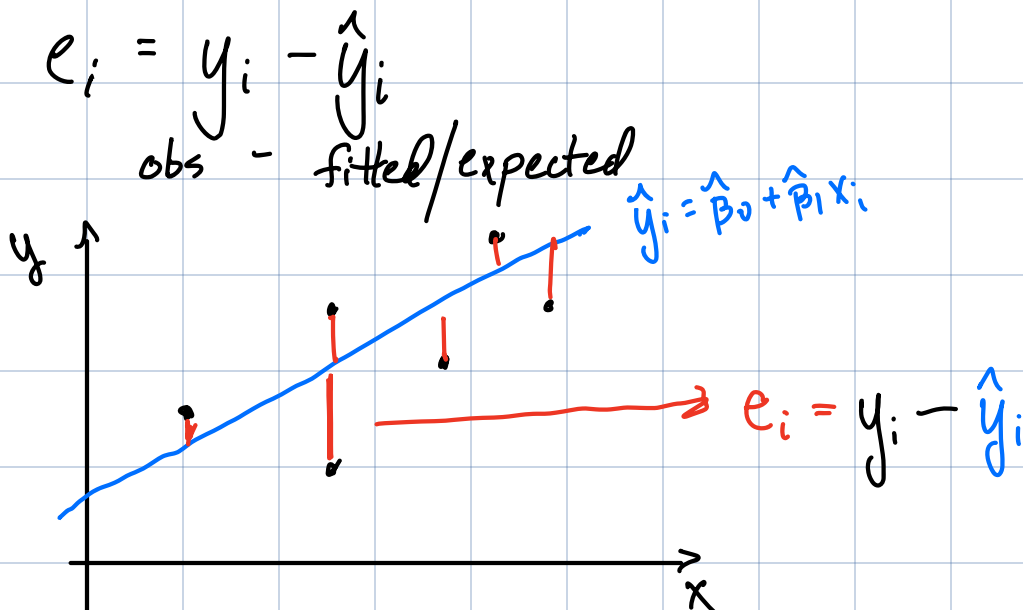
$$\sigma^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX} \right)$$

$$\hat{y}_i \sim N \left(\beta_0 + \beta_1 x_i, \sigma^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX} \right) \right)$$

What if $x_i = 0$?

What if $x_i = \bar{x}$?

II Residuals



Difference between ε_i + e_i

$$y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \leftarrow \text{don't know this}$$

$$\Rightarrow \varepsilon_i = y_i - \beta_0 - \beta_1 X_i$$

$$e_i = y_i - \hat{y}_i \leftarrow \text{we know these}$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

What is the dist'n of e_i ?

$$e_i \sim N\left(\underset{\substack{\uparrow \\ \text{Heu}}}{0}, \sigma^2 \left(1 - \frac{1}{n} - \frac{(X_i - \bar{X})^2}{SSX}\right)\right)$$

III Properties of $\hat{y}_i + e_i$

$$1. \sum_i e_i = 0$$

$$\text{Recall } \frac{\partial Q}{\partial \beta_0} = 0$$

$$\frac{\partial}{\partial \beta_0} \sum_i (y_i - \beta_0 - \beta_1 x_i)^2 = -2 \sum_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

plug in $\hat{\beta}_0 + \hat{\beta}_1$

$$\sum_i y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = 0$$

$$= \sum_i y_i - \hat{y}_i = \sum_i e_i = 0$$

$$2. \sum_i y_i = \sum_i \hat{y}_i$$

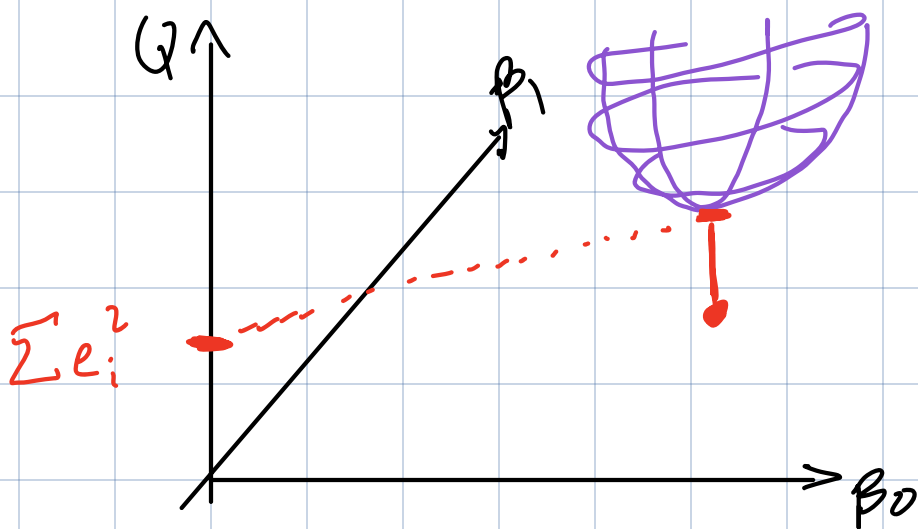
$$e_i = y_i - \hat{y}_i$$

$$= \hat{y}_i = y_i - e_i$$

$$\sum_i \hat{y}_i = \sum_i y_i - \cancel{\sum_i e_i}^0$$

3. $\sum_i e_i^2$ is the global minimum of $Q(\beta_0, \beta_1)$

$$\text{where } Q(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$



$$4. \sum_i e_i x_i = 0 \quad \text{because} \quad \frac{\partial Q}{\partial \beta_1} = 0 \quad \text{at} \quad (\hat{\beta}_0, \hat{\beta}_1)$$

$$\cancel{-2} \sum_i (y_i - \beta_0 - \beta_1 x_i) x_i = 0$$

$$5. \sum_i e_i \hat{y}_i = 0$$

$$\sum_i e_i (\hat{\beta}_0 + \hat{\beta}_1 x_i) = \sum_i e_i \hat{\beta}_0 + \sum_i e_i \hat{\beta}_1 x_i$$

$$= \hat{\beta}_0 \cancel{\sum_i e_i} + \hat{\beta}_1 \cancel{\sum_i e_i x_i}$$

IV Estimator for σ^2

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma^2)$$

Let's find an unbiased estimator, $\hat{\sigma}^2$, for σ

$$\Rightarrow E(\hat{\sigma}^2) = \sigma^2$$

Given to you: $\frac{\sum_i e_i^2}{\sigma^2} \sim \chi_{n-p}^2$

where $p = \#$ of β 's in model

SLR: $p=2$



$$V = \sum_{i=1}^n z_i^2 \sim \chi_n^2$$

$$z_i \sim N(0,1)$$

Properties of χ_n^2

$$E(V) = n$$

$$\text{Var}(V) = 2n$$

$$V = \frac{\sum_i e_i^2}{\sigma^2} \sim \chi_{n-2}^2$$

$$E(V) = n - 2$$

$$E\left(\frac{\sum_i e_i^2}{\sigma^2}\right) = n - 2$$

$$\Rightarrow \frac{1}{\sigma^2} E\left(\sum_i e_i^2\right) = n - 2$$

$$\Rightarrow \frac{1}{n-2} E\left(\sum_i e_i^2\right) = \sigma^2$$

$$\Rightarrow E\left(\frac{\sum_i e_i^2}{n-2}\right) = \sigma^2$$

↑
let this be $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{\sum_i e_i^2}{n-2} \quad \Leftarrow \text{Mean squared error (MSE)}$$

Note: MLE for σ^2 is biased $\frac{\sum e_i^2}{n}$,
but maximizes likelihood

V Hypothesis Testing & Confidence Intervals For SLR

Let's say we want to test specific values
of the slope.

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{SSX}\right)$$

Hyp. testing framework:

Hypotheses

$$H_0: \beta_1 = \beta_1^*$$

$$H_1: \beta_1 \neq \beta_1^*$$

Test Statistic

$$Z = \frac{\hat{\beta}_1 - \beta_1^*}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1^*}{\sqrt{\frac{\sigma^2}{SSX}}} = \frac{\hat{\beta}_1 - \beta_1^*}{\frac{\sigma}{\sqrt{SSX}}} \sim N(0, 1)$$

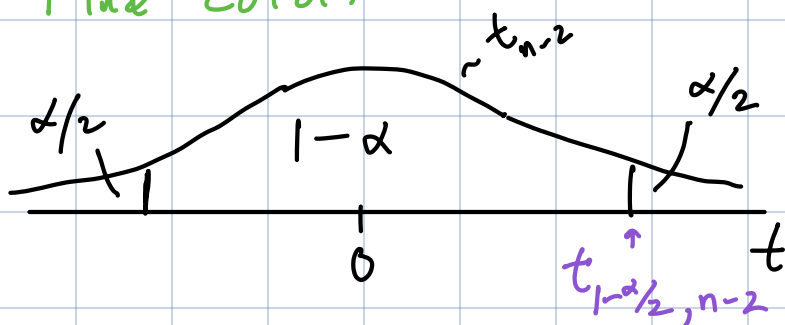
$SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$

But, σ is unknown, so let's use $\widehat{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{SSX}}$

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{\frac{\hat{\sigma}}{\sqrt{SSX}}} \sim t_{n-2}$$

Choose α

Find cutoff



Make decision

If $|t| > t_{1-\alpha/2, n-2}$, then reject H_0 ,

there is enough evidence to suggest

$$\hat{\beta}_1 \neq \beta_1^*$$

Note: Usually we use $H_0: \hat{\beta}_1 = 0$,

In this case, if we reject H_0 we

say "X is a significant predictor of y."

For the intercept

$$H_0: \beta_0 = \beta_0^*$$

$$H_1: \beta_0 \neq \beta_0^*$$

$$t = \frac{\hat{\beta}_0 - \beta_0^*}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{SSX}}}$$

Confidence Intervals

$$\text{est} \pm \text{statistic} \times \text{SE}(\text{estimate})$$

of std errors

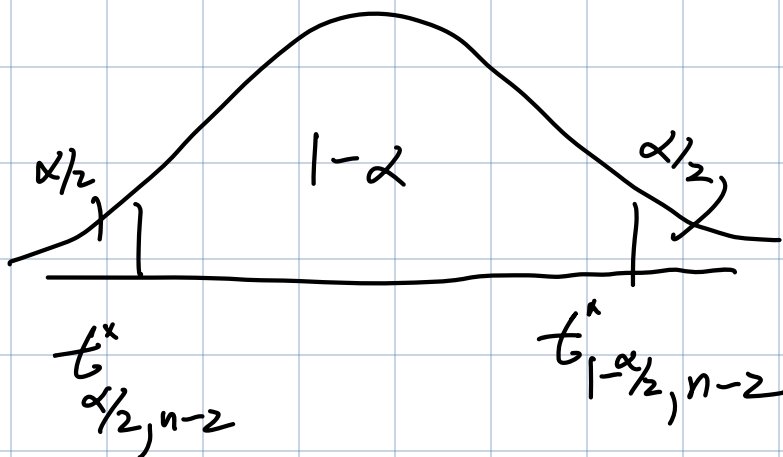
$$\hat{\beta}_1 \pm t_{1-\alpha/2, n-2} \frac{\hat{\sigma}}{\sqrt{SSX}}$$

$$\hat{\beta}_0 \pm t_{1-\alpha/2, n-2} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{SSX}}$$

But why does this work?

$$\frac{\hat{\beta}_1 - \beta_1}{\hat{SE}(\hat{\beta}_1)} \sim t_{n-2}$$

← true β_1



$$P\left(t_{\alpha/2, n-2}^* < t < t_{1-\alpha/2, n-2}^*\right) = 1 - \alpha$$

$$P\left(t_{\alpha/2, n-2}^* < \frac{\hat{\beta}_1 - \beta_1}{\hat{SE}(\hat{\beta}_1)} < t_{1-\alpha/2, n-2}^*\right) = 1 - \alpha$$

⋮

$$P\left(\hat{\beta}_1 + t_{\alpha/2, n-2}^* \hat{SE}(\hat{\beta}_1) < \beta_1 < \hat{\beta}_1 + t_{1-\alpha/2, n-2}^* \hat{SE}(\hat{\beta}_1)\right) = 1-\alpha$$

↑
this is just $-t_{1-\alpha/2, n-2}^*$

$$P\left(\hat{\beta}_1 - t_{1-\alpha/2, n-2}^* \hat{SE}(\hat{\beta}_1) < \beta_1 < \hat{\beta}_1 + t_{1-\alpha/2, n-2}^* \hat{SE}(\hat{\beta}_1)\right) = 1-\alpha$$

this is the C.I. form

probability that true β_1 is in the interval is $1-\alpha$